ShadowGen defense notes

Krengel & Geng, 20 min talk + 10 min Q&A

Numbers to never fumble

Setup: 5 logs (15k to 2.5M events), 8 configurations per log = 40 (log, miner) cells; 800 per-cell result files across methods and references; 1 h budget per cell.

Defaults: N=6, τ=5, 1,000 traces, MLE weighting, seed 42. Shipped: K=1 (single draw). Benchmark: K=5 for error bars.

Headline: Pearson 0.986–0.999 on every log; MAE 0.006–0.043; spread 0.61–0.76; flower exactly 1.0 everywhere; 3 s median per model (K=1), 14.9 s at K=5; K5/K1 ratio 5.0×.

Ranking: perfect on L1/L2/L4; one adjacent swap on L3 (pair 0.0004 apart in R1) and L5 (the two Alphas, both bottom).

PM4Py: Pearson −0.35 (L1), +0.41 (L2), −0.65 (L3), −0.53 (L4), −0.33 (L5); spread ≤0.14; flower 0.88–0.98. Ranks the two Alphas (ground truth's bottom pair) on top.

M6adapted: co-leads, MAE 0.006–0.034, flower 1.0, 36.9 s median (2.5× ours).

R3 random floor: Spearman 0.94–1.0 (!) but MAE 0.116–0.281. Why four criteria, not ranking alone.

Generator premise: exact matches of held-out variants: L5 25.9%, L2 16.6%, L3 8.6%, L1 1.1%, L4 0.09% (depth); random ≤0.01% on the five, ≤2.2% over 21 logs; ShadowGen ≥ random on all 21 (strict on 19). Within 3 edits: 44/91/47/2/80%.

Stability: per-cell std ≤0.008 (median 0.001); every single draw: Pearson ≥0.982, identical ranking; draw-to-draw Pearson std ≤0.003.

Choosing N: MAE plateau for $N \geq 4$ (five-log mean 0.020–0.021); N=6 optimal on L1, within 0.004 on L2–L4, inside noise; L5 (len 6.3) prefers shallow. Mutation rate rises monotonically with N (no peak). τ: MAE moves ≤0.007 across $\{2, 5, 10\}$.

The five logs

- **L1 Sepsis**: 1,050 cases, 15k events, 846 variants, len 14.5, TLRA 0.19
- **L2 BPI13 Incidents**: 7,554 / 66k / 1,511 / 8.7 / 0.80
- **L3 BPI17**: 31,509 / 1.2M / 15,930 / 38.2 / 0.49
- **L4 BPI18**: 43,809 / 2.5M / 28,457 / 57.4 / 0.35
- **L5 BPI19**: 251,734 / 1.6M / 11,973 / 6.3 / 0.95

TLRA $= 1-|V(L)|/|L|$: share of cases whose variant repeats. High = repetitive log, low = many unique variants.

Algorithm, in plain words

Pipeline

Log → N-gram statistics (orders 1..6, from variants, weighted by case counts) → stochastic walker generates 1,000 traces → deduplicate against the log's variants (only unseen traces stay) → token replay on the model → mean graded fitness = Gen_shadow. **Katz backoff (used twice)**

Predict from the longest recent context with enough evidence: use the deepest order $n \leq 6$ whose successor count is $\geq \tau$=5; otherwise shorten. Second use: when a mutation fires, back off one level further to find a candidate activity unseen in the resolved context but attested in a shorter one (novel, but grounded). **Good–Turing**

Estimate the chance of something never seen from how often you saw things exactly once: $p_{unseen}(s)$ = singleton successors / total successor count. A context with many one-off successors is likely to produce yet another new one. This sets how often the walker inserts a novel event; it is measured from the log, never tuned. **Termination**

$p_{end}(s) = E(s)/T(s)$: how often this context ended a trace in the log, resolved with the same backoff. Context-aware, so early "end-like" activities do not truncate traces. **Two weightings**

MLE (default): sample successors proportional to observed frequency, shadow log mirrors the expected future. Log-weighting: $\ln(c+1)$ flattens the distribution, a rare-behavior stress test. Weighting never changes the mutation rate. **Dedup + K**

Any generated trace equal to an observed variant is regenerated (up to a cap), so 100% of the shadow log is unseen. K independent draws are averaged; K=1 is the shipped default (see stability). **The two replay readings**

Token replay gives (a) graded trace fitness in $[0, 1]$ with partial credit and (b) a boolean perfect-replay flag. The metric is the graded mean. The strict fraction (Gen_accept) is a harness diagnostic: it anchors trace=0, flower=1 and exposes partial-credit inflation (trace model: 0.57 graded, 0.00 strict).

What is an alignment? (and replay vs alignment)

An alignment matches a trace against a model execution step by step, allowing mismatches at a cost (skip a log event / skip a model step); the cheapest alignment says how close the trace is to something the model can actually do. Cost zero = the model can replay the trace exactly (true language membership). Alignments are exact but expensive, and undefined on unsound nets (the tool refuses Alpha+). Token replay just fires transitions and counts missing/remaining tokens: fast, graded, slightly lenient. We measured the leniency: vs cost-zero alignments the perfect-replay flag agrees exactly on 3 of 4 checked net classes; on Inductive-infrequent agreement 0.804, all 196 disagreements one-directional over-accepts (upper bound). And we recomputed R1 with alignments: ranking unchanged (Spearman 1.0, Pearson 0.96).

The other metrics, one breath each

M2 PM4Py (structural counting): a model is "general" if its parts are visited often when replaying the log. Never sees unseen behavior; punishes rarely visited structure. Result: anti-correlated on 4/5, flower 0.88–0.98.

M3 entropic relevance: bits needed to encode the log under a stochastic model; a fitness+precision blend. Needs a stochastic automaton a Petri net does not have. DFG approximation runs everywhere but is unstable (Pearson −0.80..+0.95); proper reachability conversion builds on 24/40 cells and fails the litmus (flower mid-field, 52 bits vs 22–62).

M4 anti-alignments: find model runs maximally different from the log (the most deviant behavior the model allows); derive precision and generalization from that distance. Search is exponential.

M5 AVATAR (GAN): train a sequence GAN on the log to imitate the system, then score the model by an F-measure of fitness and precision against GAN samples. Closest paradigm to ours, but precision is folded in (flower 0.33 L1 / 0.76 L2) and it costs 9–12 h GPU training per log.

M6 bootstrap: resample the log and breed recombined traces (crossover); measure model–system recall. Same construct as ours, but recombinations only, no novel events. The shipped Entropia `-bgen` tool reports an F1 that re-adds precision: flower 0.05–0.53, MAE 0.25–0.76 (= M6original). Scored recall-only (our M6adapted), it passes and co-leads.

M7 SpeciAL (species estimation): ecology transfer: activities/N-grams are species; estimate how complete a log's coverage of the process is; we use the coverage ratio of model-simulated vs original log. Diversity is not acceptance: Pearson $-0.24..0.76$, trace pole scored 1.0 (wrong pole), flower 0.76–0.94.

M9 negative events: for each prefix, list the activities that did *not* follow it in the log (artificial negatives), each with a weight w = how confidently the log rules it out. Precision punishes a model for *allowing* high-w negatives; generalization $g_B^w = AG/(AG{+}DG)$ *rewards* it for allowing the low-w ones (impact $1-w$). So generalization is precision's complement on the same events: a permissiveness score, not acceptance of plausible future behavior. Anti-correlated on L1 (-0.55, worse than PM4Py), scores everything 1.0 on L2/L5 (incl. the trace pole), infeasible at depth. Careful: it does *not* score how well a model rejects negatives, that is its precision half.

R1 / R2 / R3 / R1-accept: 5-fold variant cross-validation fitness, 3 shuffles (ground truth); leave-one-variant-out (granularity check, same verdicts); uniformly random replay floor (must be beaten); fraction of held-out real traces replayed perfectly (acceptance ground truth).

Feasibility evidence (times, failures)

M4: scores 3/8 L1 and 3/8 L2 cells, never >2 of the 6 real miners, nothing at scale. L1: Inductive-infrequent aborts in the solver (index overflow); Alpha+ reaches 1% of traces in the hour; Heuristics 87% at the kill (nearest miss); Inductive-strict 0.1%; flower never returns. L4 does not finish the alignment step within 1 h.

M8: 3/8 L2 cells in seconds, null-alignment crash on the other five, nothing on L1/L3/L4/L5. No progress counter (wall-clock kills). Trace pole 0.90 = memorizing model read as general.

M9: L3: 2/6 miners within the hour; L4: none (trace depth 57). Nondeterministic on silent-transition nets (std 0.07).

M5 AVATAR: measured GPU training 11.8 h (L1) and 8.6 h (L2) full config (100 pre-epochs, 5000 adv steps); reduced CPU anchor 88 min, projecting 3–4 days/log (400+ CPU-h); L3 spent its full CPU hour inside pretraining; L4 OOM at 16 and 24 GB; sequence cap 20 events < L3/L4 trace lengths. In configs: `runtime_s` is sampling only, `training_time` is the real cost.

-bgen at scale: 6–10 min per L4 cell, 5/8 within the hour, 3 fail inside the tool. Published tool times out on L4 (2.5M events).

SpeciAL at scale: crashes on L5's 1.6M-event log for Alpha/Alpha+ (6/8 cells return); version skew: PyPI 0.1.3 on L5 vs authors' source on L1–L4 (disclosed).

Kept overruns (disclosed): 5 L4 cells kept despite exceeding 1 h: PM4Py Ind-strict 3.5 h; 1-gram ablation Ind-strict 4.0 h; SpeciAL Heur-strict 2.2 h, Heur 1.2 h; R3 Ind-strict 1.2 h. Sentineling them would only delete cells from baselines that already fail.

References cost: R1 worst cell 5.7 h; LOVO 46 h; R2 on L4 Ind-strict sampled 50 of 28,457 variants.

Likely questions, short answers

Why must the flower score 1.0? Construct: generalization = acceptance of future valid behavior, nothing else. The flower accepts everything, so its generalization is maximal by definition; its uselessness is a precision and simplicity verdict. Scoring it <1 proves the metric mixes something else in.

Is the shadow log circular (built from the same log)? The generator sees only the log; the model enters at replay only. And the premise is tested: trained on 4/5 of the variants, up to 26% of generated traces exactly match held-out variants the generator never saw; random ≈0. Dedup + disjoint folds rule out memorization.

But ShadowGen cannot label a generated trace valid or invalid? Correct, and it never claims to. The four-dimension definition (slide 2) states the construct: accept future valid behavior. ShadowGen is an estimator of it, probing with plausible traces. Three measured answers. (1) The estimate is validated against a ground truth whose test traces are provably valid, real held-out variants, and it tracks that at Pearson 0.986–0.999. (2) The premise is measured: up to 25.9% of generated traces are exact real held-out variants (random ≈0), and even the context-novel traces land within 3 edits of one 13–57% of the time (random floor 4.6%). (3) If the shadow log were invalid noise, the score would degenerate into permissiveness, which is exactly the random floor R3: MAE 0.28 against our 0.023. The report states the bound itself (threats): a shadow trace is plausible by construction where an R1 test trace is provably valid.

Why token replay, not alignments? Alignments refuse unsound nets (Alpha+); replay is graded and fast. We measured the replay leniency (0.804 agreement, one-directional) and re-verified R1 with alignments (ranking holds).

Why K=1? Measured single-draw stability: std ≤0.008, each draw alone keeps Pearson ≥0.982 and the identical ranking; seeded, so deterministic in use. K=5 exists for error bars.

Why N=6 and not per-log tuning? Calibration is a plateau ($N \geq 4$); N=6 is optimal or within 0.004 everywhere except short-trace L5; differences sit inside sampling noise. The mutation rate is monotone in N, so novelty offers no optimum; fixing one N keeps the metric parameter-free in use.

Isn't 1 hour arbitrary? It is a practitioner-scale budget, uniform for everyone, with sentinels carrying measured evidence; runtime is a first-class outcome. Five kept overruns are disclosed, all from baselines that already fail.

Why does PM4Py fail? It rewards frequently visited structure. Permissive models spread visits thin; restrictive ones concentrate them. That is structure, not future-behavior acceptance: spread 0.09, Alphas ranked on top.

Difference to bootstrap (closest rival)? Same construct, and it co-leads when scored recall-only. But it only recombines observed material (no novel events), costs 2.5× more, and its shipped F1 reading is precision-contaminated (the four-scorings study shows the precision component alone destroys calibration).

Where is the generalization in M9 (negative events)? The intent is close to ours: a weakly ruled-out negative (low w) is behavior the log cannot confidently call invalid, so a general model should accept it, and g_B^w rewards exactly that. The flaw is that the impact on generalization is $1-w$ where w is the impact on *precision*, over the *same* negative events, so the two are algebraically locked. The authors say so themselves (usefulness for generalization is inversely related to its use for precision) and recommend reading generalization with lower priority. What is left is a permissiveness score, and permissiveness is the bar our random floor R3 already clears. Key difference to ShadowGen: our probes are *generated* to look like real future behavior; M9's probes are everything not confidently ruled out, most of which is genuinely invalid, so rewarding acceptance of it rewards underfitting. L1 evidence: Heuristics 0.11 and Inductive-strict 0.19 (the two best generalizers per R1) score *lowest*; the memorizing trace model 0.48; flower 1.00. On L2/L5 it collapses to 1.0 for every model (negatives following a force-fired event are skipped, so almost nothing is evaluated). We run CoBeFra's `generalization` field, not its precision, so the critique is on the right number. Paper: IEEE TKDE 26(8):1877–1889, 2014, doi 10.1109/TKDE.2013.130; free PDF at lirias.kuleuven.be.

Isn't M9's trace pole (0.48 on L1) better than yours (0.57)? Correct on the numbers, and the report does not fault M9 for the L1 value: every graded (partial-credit) reading inflates the trace pole, the ground truth included (R1 graded reads 0.641 on that model). That is exactly why the benchmark carries the strict acceptance reading, which anchors the pole: R1-accept 0.000 and Gen_accept 0.000 on L1. The M9 criticism rests on what partial credit cannot explain: the ordering inversion on L1 (memorizer 0.48 above Heuristics 0.11 and Inductive-strict 0.19) and the L2/L5 collapse, where the trace pole reads a full 1.0. M9 has no strict counterpart reading, so its inflation cannot even be diagnosed.

How trustworthy is the ground truth? Held-out real variants are provably valid behavior. Robust to granularity (R2 gives the same verdicts) and to the engine (alignment recomputation keeps the ranking). Both are bounded by log representativeness, which we span deliberately (TLRA 0.19–0.95).

Does representativeness break the metric? Registered prediction said calibration error grows as TLRA falls. Refuted: best MAE at TLRA 0.49, worst at 0.95. TLRA governs exact-matchability of the future, not calibration.

What about precision overall? Deliberately out of the score. The metric is to be read next to a precision metric; the quadrant stays diagnostic.

Limitations to volunteer: control-flow only (no data attributes); dependencies beyond order N not modeled; strict reading saturates at depth (L4); token-replay core shared with R1 (checked); 1-gram suffices on short repetitive logs (L5).

Talk cues (12 slides, about 100 s each)

1. Title: one sentence on the claim: a validated, affordable generalization metric.
2. Four dimensions; generalization = least validated; construct: future valid behavior only.
3. Gap: 8 published metrics, 0 validated on real logs; what ShadowGen fills.
4. Test bed: permissiveness spectrum, trace to flower; the litmus (flower must be 1.00).
5. Pipeline: log to N-grams to shadow log to graded replay; model only at replay.
6. Two backoffs: resolve (deepest supported), then invent (one level further); rate from Good–Turing.
7. Validation: 5 logs, R1 cross-validation, four criteria (ranking alone too weak), 1 h budget.
8. Tracks ground truth: 0.99+, 3 s, MAE 0.006–0.043; single-draw stable.
9. Shadow log is real future behavior: exact matches of held-out variants; 21-log sweep.
10. PM4Py does not measure generalization: anti-correlated, litmus failed; mention M9/M3 construct drift.
11. Accurate and affordable: pareto; alone in the accurate corner with bootstrap; AVATAR 9–12 h GPU.
12. Conclusion: validated, affordable, interpretable; construct + litmus as reusable contributions.

If the supervisor pushes on rigor

Self-correction story: our first N=6 justification (mutation-rate peak on one log) did not reproduce when re-measured; we retired it, swept all five logs (MAE) and all 21 (rate), and re-justified by calibration. The value survived, the criterion changed.

Preregistration: the TLRA prediction was registered, tested, and refuted; reported as a negative result.

Provenance: every cell writes a sidecar JSON (params, raw iterations, runtime, integrity counters); figures and tables regenerate from those files; a result without a config is invalid.